

QUARTERLY OPERATIONS REVIEW

Safety, Equipment & Supply Chain Performance

Presented to the Senior Management Team • Weeks 6–8

BLUF: Demand forecasting and inventory policy work this quarter cut projected excess buffer stock by ~86% and identified a lower-cost distribution plan saving an estimated \$172K/year — while a formal reorder policy reduces stockout risk from a coin-flip to 1-in-20.

Three headlines for this quarter

5.3%

Forecast error (MAPE)

45-day demand forecast, validated on held-out history

95%

Target service level

Reorder policy now sized to a defined stockout-risk tolerance, not guesswork

\$172K

Est. annual savings

From optimized warehouse-to-region distribution routing

What this means for the business

- Inventory is now sized by statistics, not intuition — freeing working capital previously tied up in oversized buffers.
- A data-backed reorder policy quantifies and controls stockout risk at a Board-approved service level.
- Distribution routing is now cost-optimized rather than habit-based, with savings that compound every week.

Workplace Safety Performance

Placeholder — replace with your actual Week 6 safety analysis before presenting.

Metric	Prior Quarter	This Quarter	Trend
Recordable incident rate	[value]	[value]	[↓ improving]
Near-miss reports filed	[value]	[value]	[↑ more reporting]
Safety training completion	[value]	[value]	[on target]
Days since last lost-time incident	[value]	[value]	[value]

Key insight from Week 6

[Insert your one-sentence takeaway from the Week 6 safety analysis here — e.g. which intervention drove the biggest improvement.]

Equipment Health & Reliability

Placeholder — replace with your actual Week 7 equipment-health analysis before presenting.

[_ %]

Fleet uptime

[_ days]

Mean time between failures

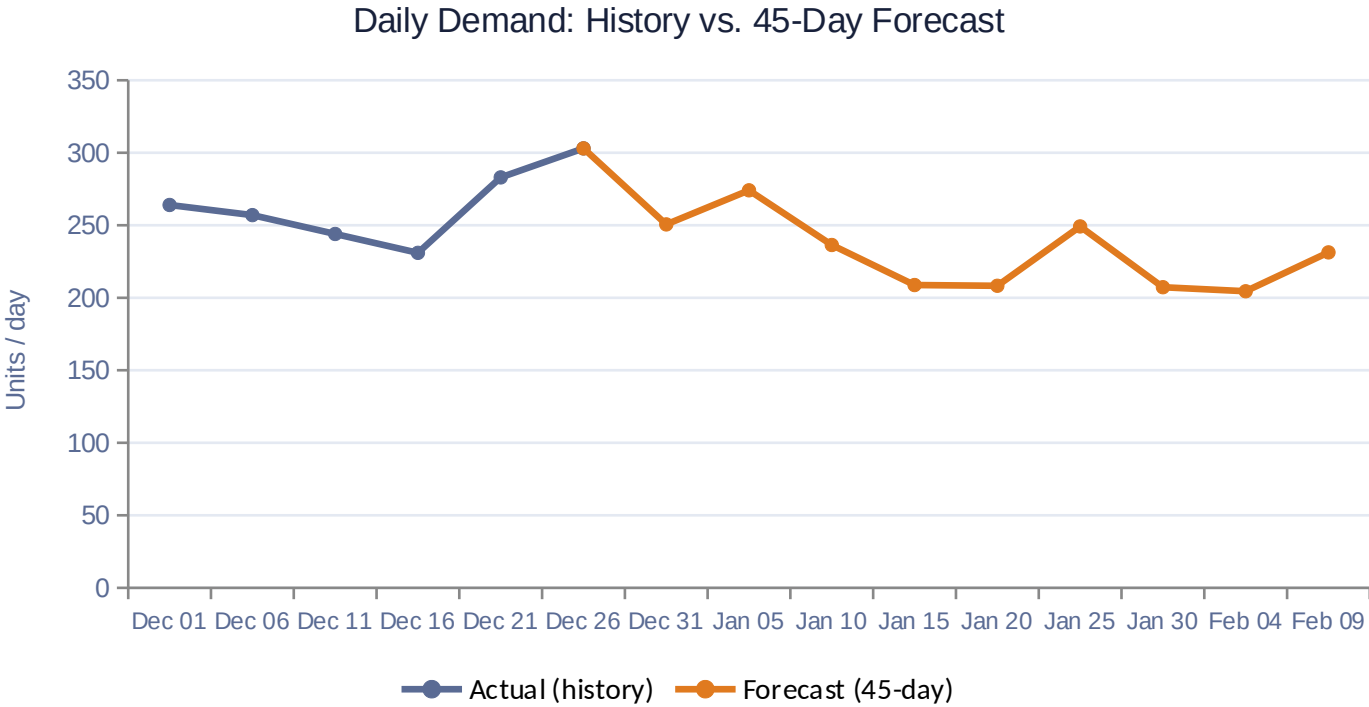
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Predictive-maintenance flags resolved

Key insight from Week 7

[Insert your one-sentence takeaway from the Week 7 equipment-health analysis here — e.g. which asset class needs earliest replacement or which maintenance change had the biggest impact.]

Demand Is Forecastable — and Now Forecast



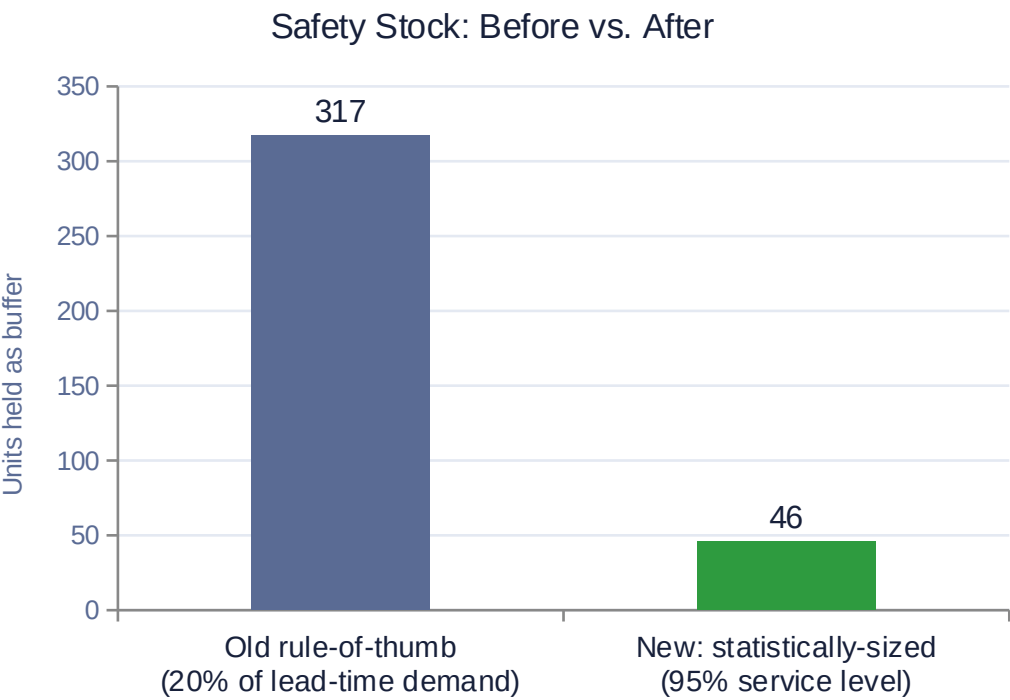
5.3% MAPE on holdout test window

±15.4 RMSE (units/day)

45 days Forecast horizon produced

Model includes a promotions regressor and Kenyan public holidays as explicit external drivers of demand.

Right-Sizing the Safety Buffer



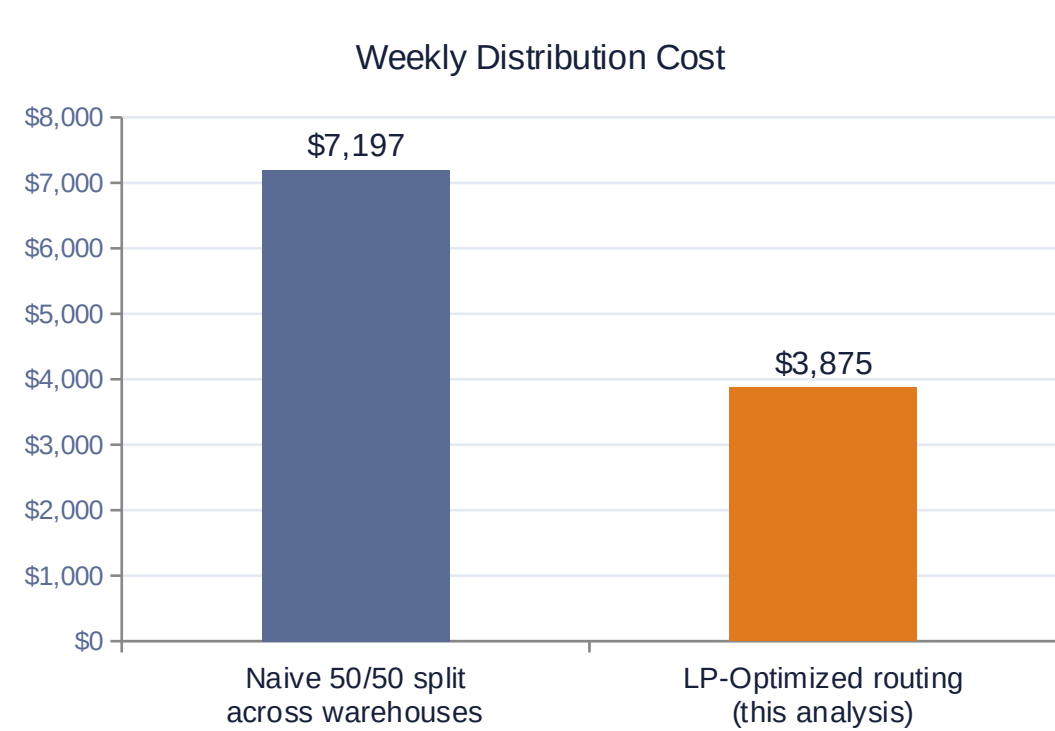
86%

reduction in units tied up as excess buffer stock — freeing capacity and working capital, at the same 95% protection level.

Assumption	Value
Lead time	7 days
Target service level	95% (Z = 1.645)
Demand variability (σ)	10.5 units/day — measured from forecast error, not assumed
Reorder point	≈ 1,633 units

Without a sized buffer, the reorder point sits exactly at average demand — a coin-flip (~50%) chance of stockout every cycle. Sizing it to 1.645σ cuts that risk to 5%.

Lowest-Cost Path From Warehouse to Region



46% lower weekly shipping cost — about \$3,320/week, or an estimated \$172K/year at current demand levels.

Warehouse → Region	Units/week
Nairobi DC → Central	923
Mombasa DC → Coast	554
Nairobi DC → Western	369

Solved via linear programming (PuLP): minimize total shipping cost subject to each region's demand being met and each warehouse's weekly capacity.

Three Moves, Ranked by Impact

1

Adopt the LP-optimized distribution routing

Shift standing shipping instructions to the cost-minimizing warehouse-to-region routes identified this quarter.

ROI: Est. \$172K/year in avoided shipping cost (46% weekly reduction), with no change to service levels.

2

Formalize the statistical reorder-point policy

Replace ad-hoc reordering with the calculated reorder point and safety stock, refreshed each forecast cycle.

ROI: Cuts stockout risk per cycle from ~50% to 5%, while cutting buffer inventory ~86% — freeing working capital currently sitting idle as excess stock.

3

Operationalize the demand forecast into planning cycles

Feed the 45-day Prophet forecast (5.3% MAPE) directly into procurement and staffing plans, refreshed weekly.

ROI: Tighter, evidence-based planning cycles reduce both rush-order premiums and excess-stock write-offs; exact \$ impact to be quantified with Finance once per-unit holding/rush-order costs are shared.